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ORIGINAL ARTICLE

Control of a dual stage magnetostrictive actuator and linear motor feed drive system

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Abstract A dual stage feed drive system is well suited to satisfying current demands of high performance machining with tight position control under high feed rates in the presence of disturbances. It does this by integrating an actuator with high position resolution and fast response together with a conventional linear drive. A magnetostrictive actuator (MA) with a bandwidth in the kHz range capable of several kN of force output is an ideal candidate for use as the fine positioning element in such a dual stage system. However, MAs display significant hysteresis in their performance. This makes the effective implementation of real-time fast servo control challenging. In this paper, a dynamic Preisach model is utilized to reduce the undesired nonlinearity. A sliding mode controller (SMC) is designed to deal with uncertainties such as the Preisach modeling error as well as external disturbances so as to ensure a robust stable system. Experimental results show the servo performance improvement during a feed step test for single axis control and a single axis component of a sharp path interpolation test.

Keywords Dual stage actuation (DSA) · Fast servo control · High speed machining · Magnetostrictive

1 Introduction

Intensive competition in the manufacturing industry requires that firms reduce production cycle times while achieving tight tolerances at a low cost. This can best be achieved by using high performance cutting (HPC) concepts. Cutting forces in these applications, which are disturbances to the feed drive system, will behave quite differently from traditional machining processes due to the high cutting speed as well as the accompanying high temperature and high material removal rate, thus the cutting forces will have the following features: (1) large variation in cutting force magnitude (especially for cutting difficult-to-machine materials), (2) rapid variation with time while undergoing a complex contour or sculptured surface machining operation, and (3) high frequency content due to the high spindle speeds typically used.

Feed drive technology is an indispensable element of high performance cutting and it must undergo considerable improvement to meet the above-mentioned challenges. New feed drives must demonstrate especially good dynamic characteristics, i.e., control loops with small time constants and actuators that can generate high forces over short periods of time with the desired positioning accuracy [1].

Ball screw systems and linear motors each have their own advantages and disadvantages in feed drive applications. Although ball screw systems are still dominant as feed drives, their dynamics are limited, due to the nature of their mechanical transmission elements [2–4]. Linear motors generate linear motion directly and have no need for a mechanical transmission system. However, for linear motors, both contouring accuracy and rigidity are largely determined by the control loop [5–7]. Solid-state actuators are now part of the field of modern machining. They are able to provide high force and stiffness under motion with nanometer scale accuracy under high bandwidth operation. However, they are limited by their very short working stroke.

Recently, there has been increased interest in dual stage actuation (DSA) systems to increase both system bandwidth and positioning resolution. Representative examples, which adopt the dual stage actuation concept, are hard disk drives,

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macro/micro robot manipulators, and XY linear positioning tables. However, the dual stage design is rarely found in machining applications. This paper presents the design and servo control of a dual-stage actuation system, in which a linear motor (LM) is used as the coarse actuator and a magnetostrictive actuator (MA) is used as the fine actuator. This paper is organized as follows: Section 2 discusses the design of the proposed DSA system. Section 3 presents the removal of nonlinearities using the Preisach method and the subsequent SMC controller design. Section 4 shows selected experimental results of the system to step and sharp corner tests. Finally, concluding remarks are given in Section 5.

2 Design and modeling of the dual stage system

2.1 Design of the dual-stage actuation system

Figure 1 demonstrates the proposed dual-stage system configuration that makes use of the two drive elements, a course actuator followed by a fine actuator. The design utilizes a linear motor and a magnetostrictive actuator as the course and fine drive elements respectively. An additional moving linear block is added to the linear bearing to carry the table, workpiece, and corresponding fixtures. The linear motor is responsible for carrying the fine actuator, which has one end fixed with the linear motor's body while the other end is connected to the additional moving part. The fine actuator with short stroke and high stiffness is responsible for delivering high frequency and very accurate small movements. It has to carry as low a moving mass as possible in order to achieve higher acceleration. The coarse actuator carries the fine one, thus it has more moving mass, which can act as a low-pass filter for high frequency vibrations. It is responsible for delivering the table and workpiece to the vicinity of the exact reference position. While the fine actuator corrects for the tracking error made by the coarse actuator due to its inability to respond to high frequency cutting force disturbances and to make up for its lack of dynamic stiffness. See Figure 2.

2.2 Modeling of the dual-stage actuation system

A system identification method was used to model the linear motor. Preliminary study on the MA indicated that its nonlinearities would first need to be addressed before a controller could be developed for it. Thus a Preisach model (PM) was used to reduce/eliminate the inherent hysteresis in its performance. The PM has received considerable attention in the past [8–11]. A PM is purely phenomenological in nature. Essentially, it represents an attempt to describe and generalize experimental data. Unlike its counterpart, physics-based models, it can be easily solved and inversed, which makes it suitable for fast servo control applications.

The PM represents a complicated operator, which in our case is the hysteresis operator Γ, as a superposition of simple operators. In other words the PM is a decomposition of the complicated hysteresis operator Γ into a sum of simpler operators. It is similar to the process of decomposing a complicated periodical function into a sum of simpler sinusoidal components with different magnitudes and frequencies. The original PM was essentially a static rate-independent model, which was unable to describe dynamic hysteresis phenomena. By introducing dynamic effects, a standard PM can be generalized into a dynamic PM. The identification method used and the resultant PM as well as the hysteresis reduction will be discussed in a later section.

3 Servo controller design

Since there are fine and coarse actuators working together in one system, the dual stage feed drive is a DISO (dual-input/single-output) system. Thus the corresponding controller design is a special case of a multi-input/multi-output (MIMO) design.

Optimal linear-quadratic (LQ) control is one of the most commonly used control strategies to handle MIMO control systems. The corresponding controller design is usually associated with state space techniques via the solution to a Riccati equation in the time domain. It needs to construct

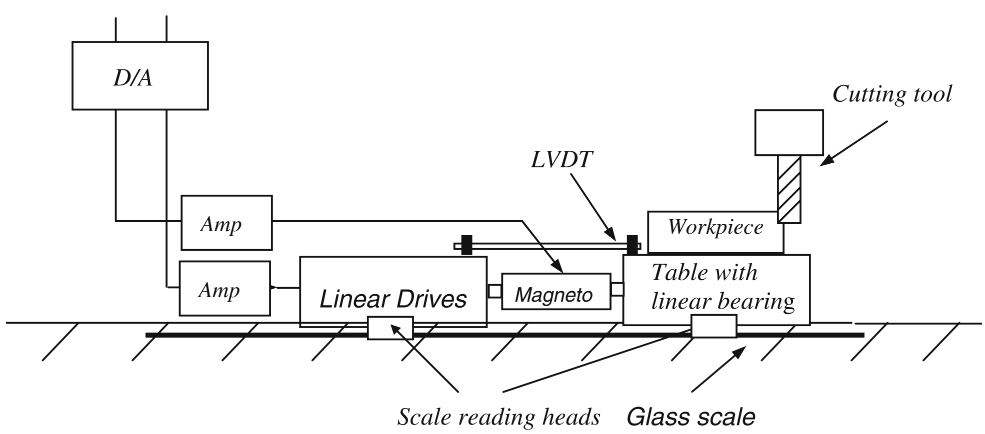


Fig. 1 Schematic layout for a proposed dual_stage feed drive system

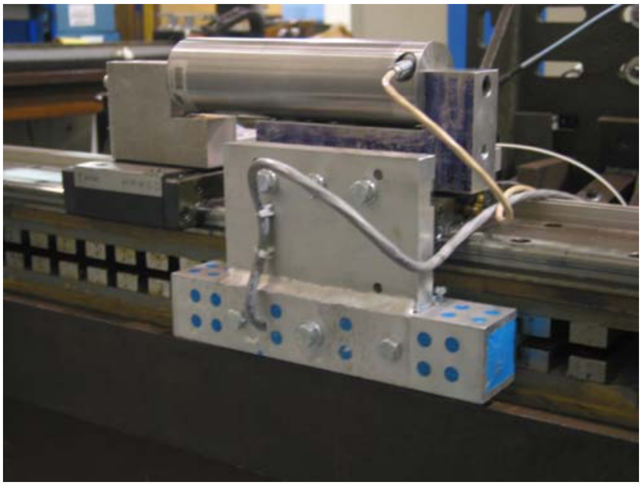


Fig. 2 Prototype of DSA system

state observers and its controller structure is obscured for the sake of the mathematical optimization procedures. Therefore, higher order controller design often results, which can be hard to implement in the case of real time fast servo control. Another LQ controller design approach via input-output transfer function has been utilized to control a dual-stage actuation system [12]. It was reported that such a time-domain design can easily handle non-square systems and there is no need to construct state observers and the controller structure is much more evident.

Efforts have been made, such as master-slave methods and PQ methods, etc. [13, 14], to treat the mentioned MIMO design as a sequence of SISO designs under the assumption that there is very little interaction between the two individual SISO loops. Unfortunately in this case considerable interaction does exist and this deteriorates the overall performance within some frequency range. Furthermore the stability of the overall system is a major concern due to the separate nature of the designs and thus it cannot be guaranteed in some cases. Finally, these methods don't represent a systematic approach in this case as they don't have the facilities to coordinate the two individual loops so that the optimal performance of the overall system can be achieved.

Magneto actuators by their nature are non-linear and contain a hysteresis loop. As the most important component of a dual-stage system the magneto actuator's performance determines the overall DSA system's performance. Therefore, in order to achieve a high degree of control performance, nonlinear modeling and control strategies are necessary. In this paper, a new approach will be proposed to control the DSA system containing a highly nonlinear component by using a combination of feedforward hysteresis cancellation and a feedback robust controller. The feedforward compensation part is based on a dynamic hysteresis model of the magneto actuator, while a sliding mode controller (SMC) is used to compensate uncertainties such as the error in hysteresis cancellation and to ensure system stability.

In order to design a SMC controller, consider the dynamic system:

$$\mathbf{X}^{(1)} = \mathbf{F}(\mathbf{x}) + \mathbf{B}\mathbf{u}$$

where u is the control input, $x = \left[x, x^{(1)}, \ \ldots x^{(n-1)}\right]^T$ is the system state vector, and x is the output of interest. Thus the ith component of the system with bounded uncertainties can be written as:

$$x_i^{(n_i)} = f_i(x) + \sum_{j=1}^{m} b_{ij}(x)u_j, \ i = 1, \ldots, m; j = 1, \ldots, m$$

$$\left|\widehat{f}_i - f_i\right| \leq F_i, \ i = 1, \ldots, m$$

$$B = (I + \Delta)\widehat{B}, \left|\Delta_{ij}\right| \leq D_{ij}, \ i = 1, \ldots, m; j = 1, \ldots, m$$

A contouring problem with respect to the machine tool's feed drive servo control can be stated as a control objective to make the state $\mathbf{x}$ track a specific time varying state $\mathbf{x_d} = \left[x_d, x_d^{(1)}, \ldots x_d^{(n-1)}\right]^T$ in the presence of model imprecision on A(x) and/or B(x) as well as external disturbances. For the specific tracking control applications on a DSA system, let $\widetilde{x} = x - x_d$ be the tracking error in the variable x, and let $\widetilde{x} = x - x_d$ be the tracking error vector. The desired system dynamics on a time-varying sliding surface is defined as $s(\mathbf{x}, t)$=0, where

$$s(\mathbf{x}, t) = \left(\frac{d}{dt} + \lambda\right)^{n-1} \widetilde{x}$$

λ is desired and achievable bandwidth of the controlled dynamic system. It is a strictly positive constant. Generally, in a practical tracking problem, the equation is used most often by selecting n=2, i.e.:

$$s = \left(x^{(1)} - x_d^{(1)}\right) + \lambda(x - x_d) = \widetilde{x}^{(1)} + \lambda\widetilde{x}$$

It is clear that the problem of tracking $\mathbf{x}=\mathbf{x}_d$ is equivalent to that of remaining on the sliding surface $s(\mathbf{x}, t)$=0 for all t> 0. In order to maintain the system on the sliding surface $s(\mathbf{x}, t)$ =0, the system must satisfy $s(\mathbf{x}, t)$=0 and more strictly, $s^{(1)}(x, t) = 0$. We can obtain:

$$s^{(1)}(\mathbf{x}, t) = \left(x^{(n)} - x_d^{(n)}\right) + \lambda\left(x^{(n-1)} - x_d^{(n-1)}\right)$$
$$= x^{(n)} - x_d^{(n)} + \lambda\widetilde{x}^{(n-1)}$$

substitute $x_i^{(n_i)} = f_i(x) + \sum_{j=1}^{m} b_{ij}(x)u_j$, or in the form of a vector, $x^{(1)} = F(x) + Bu$, we can solve the control input in the form of the vector as shown below if $f(\mathbf{x})$ is known perfectly and the model is accurate:

$$\mathbf{u} = \mathbf{B}^{-1}\left(-\mathbf{f} + \left(x_{di}^{(n)} - \lambda\widetilde{x}^{(n-1)}\right)\right)$$

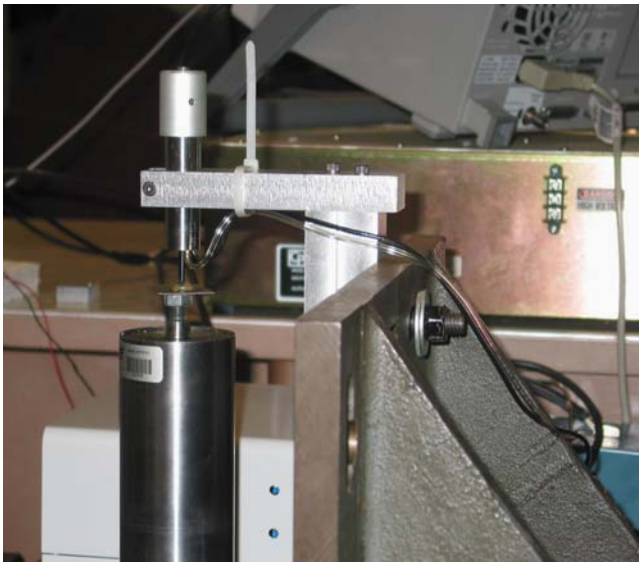


Fig. 3 Test rig for MA investigation with a LVDT used as position sensor

Due to the existing uncertainties in the system modeling as well as external disturbances, we rewrite the above equation as follows by adding a discontinuous term **k**sgn(**s**).

$$\mathbf{u} = \mathbf{B}^{-1}\left(-\widehat{f} + \left(x_{di}^{(n)} - \lambda\widetilde{x}^{(n-1)}\right) - \mathbf{k}\, sgn\,(\mathbf{s})\right)$$

The following Lyapunov function $V(t) = 1/2[s^2]$ is used to ensure that the controller law satisfies individual sliding conditions of the form :

$$\frac{1}{2}\frac{d}{dt}s_i^2 \leq -\eta_i|s_i| \quad (\eta_i > 0)$$

in the presence of parametric uncertainty.

$$\begin{aligned}\mathbf{V}^{(1)} = \mathbf{s}^{(1)}\mathbf{s} &= \Big[-\mathbf{f} + \left(x_{di}^{(n)} - \lambda\,\widetilde{x}^{(n-1)}\right) \\ &\quad -\mathbf{B}\mathbf{B}^{-1}\left(-\widehat{f} + \left(x_{di}^{(n)} - \lambda\,\widetilde{x}^{(n-1)}\right) - \mathbf{k}\,sgn\,(\mathbf{s})\right)\Big]\mathbf{s} \\ &= \left[\left(\widehat{f} - \mathbf{f}\right) - \mathbf{k}\,sgn\,(\mathbf{s})\right]\mathbf{s} \\ &= \left(\widehat{f} - \mathbf{f}\right)\mathbf{s} - \mathbf{k}|\mathbf{s}|\end{aligned}$$

Recalling $\left|\widehat{f_i} - f_i\right| \leq F_i\ , i = 1, \ldots, m$, let $k = F + \eta$, we can achieve $\frac{1}{2}\frac{d}{dt}s^2 \leq -\eta|s|$ as desired. Further, let $\mathbf{V}^{(1)} = \mathbf{s}^{(1)}\mathbf{s} = -k_S\mathbf{s}^2, k_S > 0$, is a feedback gain. So,

$$\mathbf{s}^{(1)}\mathbf{s} = \left[-\widehat{f} + \left(x_{di}^{(n)} - \lambda\,\widetilde{x}^{(n-1)}\right) - \mathbf{B}\mathbf{u} - \mathbf{k}\,sgn\,(\mathbf{s})\right]\mathbf{s} = -k_S\mathbf{s}^2$$

$$\mathbf{u}_{smc}(k) = \mathbf{B}^{-1}\left[\left(x_{di}^{(n)} - \lambda\widetilde{x}^{(n-1)}\right) - \widehat{f} + \mathbf{k}\,sgn\,(\mathbf{s}) + \mathbf{k}_S\mathbf{s}(k)\right]$$

$$s(k) = \lambda\left(x_r(k) - x(k) + \left(x_r{}^{(1)}(k) - x^{(1)}(k)\right)\right)$$

Based on the fundamental principle of constructing a disturbance observer, a simple disturbance observer introduced by [15] will be used during the control law design given by the following equation. The variation in disturbances here is considered to be constant and remains between upper (d^+) and lower (d^-) limits measured on the machine due to the very fast sampling rate and the slowly varying disturbance.

$$\widehat{d}(k) = \widehat{d}(k-1) + \rho\kappa sT$$

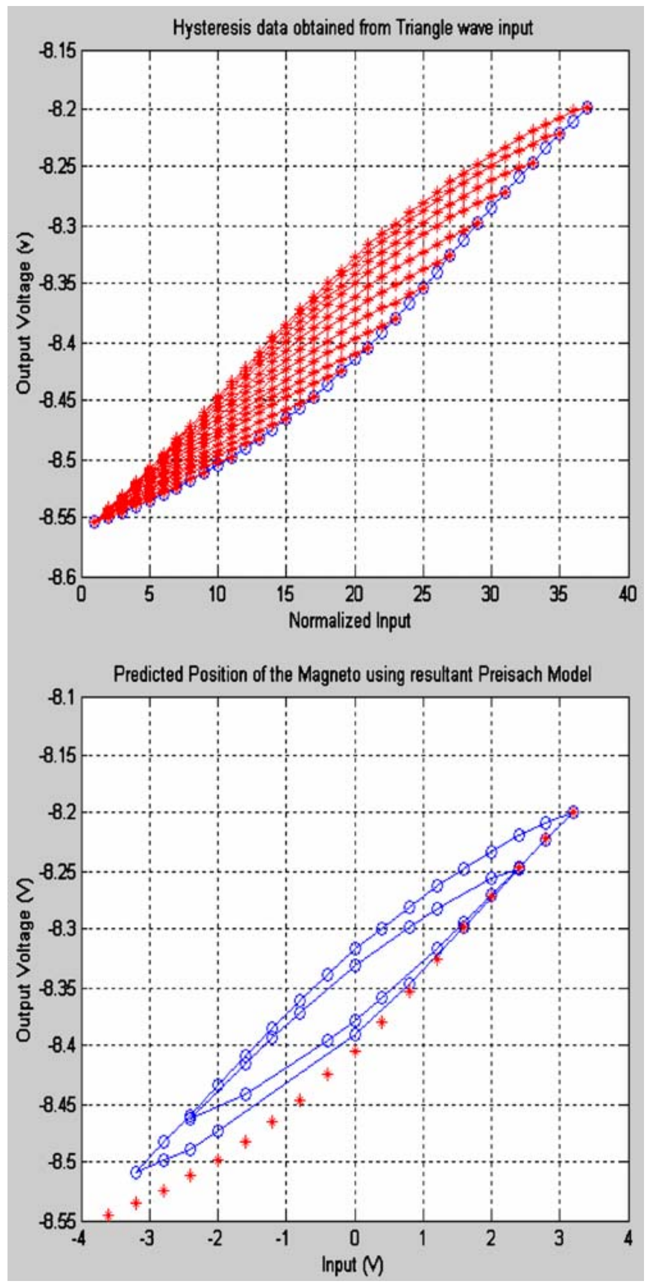


Fig. 4 Collected hysteresis data (left) and prediction using resultant standard Preisach Model(PM) (right) where 1 Hz sine wave is used as an input

where T is the sampling interval, k is the control sampling interval counter in the discrete time domain, ρ is the parameter adaptation gain, and κ is used to impose limits on the integral control action against the disturbance as,

$$\kappa = \begin{Bmatrix} 0 \; if \; \widehat{d}(k) \leq d^- \; and \; s \leq 0 \\ 0 \; if \; \widehat{d}(k) \geq d^+ \; and \; s \geq 0 \\ 1 \; other \; conditions \end{Bmatrix}$$

Hence, the estimated disturbances are always kept within the predetermined bounds, i.e., $\widehat{d}(k) \in [d^-, d^+]$.

4 Test and experimental results

In this section, we present the resultant controller and experimental results from the DSA system. An Extrema-100 magnetostrictive actuator (MA) was utilized as the fine actuator in the DSA system. The different controllers were implemented on a 650 *MHz* P4 and the position feedback signals from the linear motor and the table were collected by a linear encoder with 1 μm resolution. The I/O card consists of eight, 13-bit channels of analog input and output, and 24 configurable channels of digital I/O. The software is running on RT-Linux and a sampling rate of 10 KHz was adopted.

In order to investigate the magnetostrictive actuator's characteristics and behavior for servo control applications, a test rig was built to test it. Figure 3 shows the test rig where an LVDT is used as a position sensor. Test code has been written to obtain data in the Preisach model format. Additional code has also been written to form the Preisach model (see Fig. 4). The test rig and test code can also be used to verify the generalized Preisach model (GPM).

A test on the magneto actuator was also conducted using a sinusoid input at frequencies from 5 to 1000 Hz without load. The output displacement was recorded using a proximity sensor. The MA shows far higher bandwidth than the traditional linear drives from Fig. 5. The magnitude of the ratio between the output and the input start to drop slightly from around 450 Hz. It can also be seen noise is introduced at the high frequency zone but with a reduced magnitude. In addition, the shape of the resultant hysteresis also varies as the input frequency increases. This phenomenon also indicates that the magneto actuator's hysteresis is frequency dependent.

The standard PM was generalized (GPM) by introducing dynamic effects, namely, the up-down two position relay used in a standard PM switches from up to down at a finite rate instead of changing instantaneously. Figure 6 (left) shows that the resultant GPM models the dynamic behavior of the magneto actuator with a high degree of accuracy.

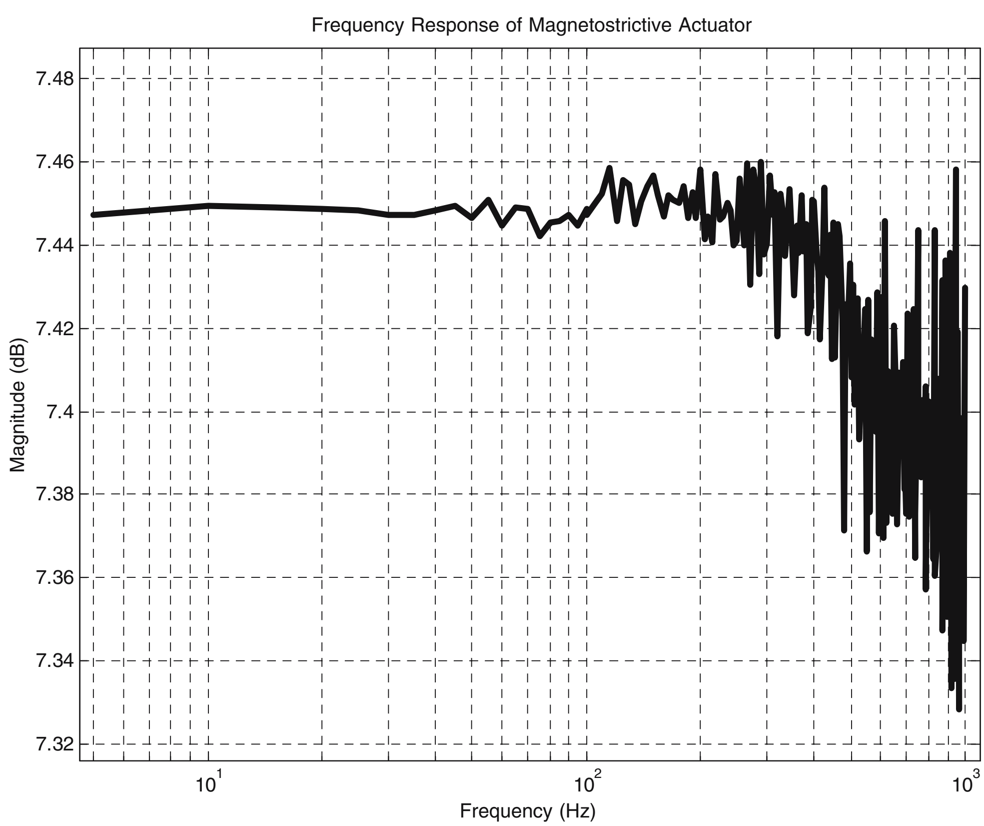


Fig. 5 Frequency response of MA from 5 to1000 Hz

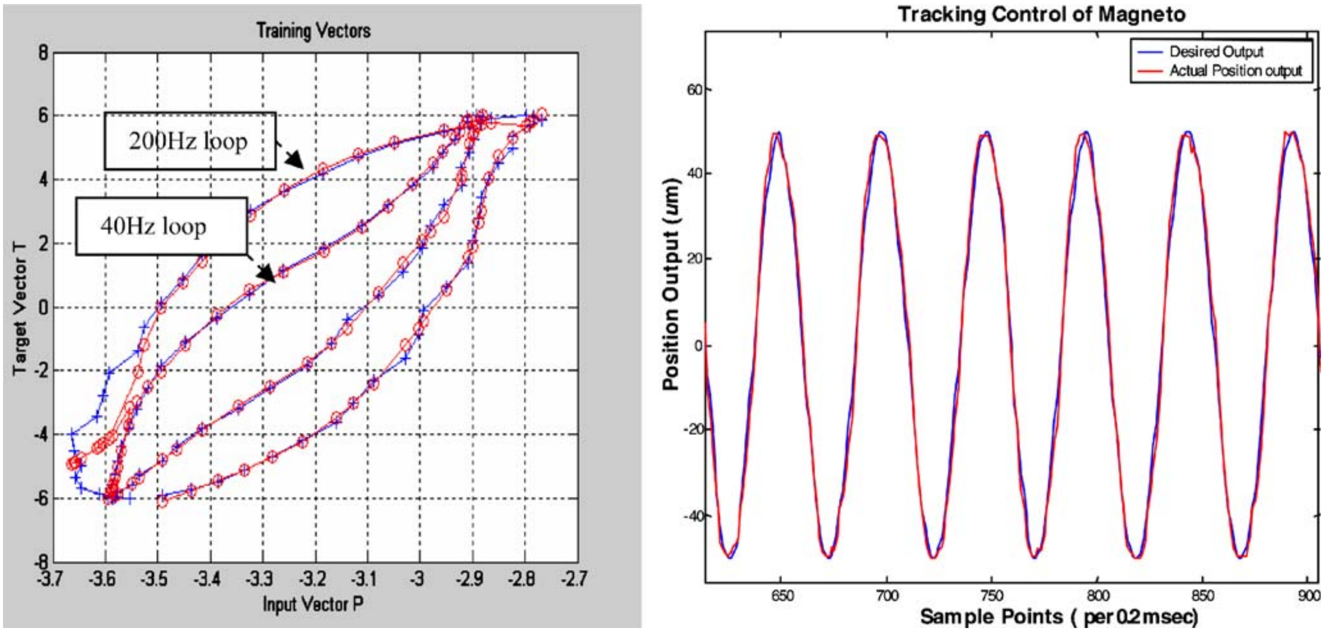


Fig. 6 Measured hysteresis at 40 and 200 Hz (+) and dynamic model prediction (o) and tracking of a sinusoidal input (right) at 100 Hz

Figure 6 (right) shows the sine wave tracking performance using a resultant generalized PM model.

The model structure of the linear motor is derived from its physical law. The plant dynamics can be described by the following transfer function. This equation relates the amplifier input voltage to the plant output, which is the actual position of the moving carriage in this case. $G_M(z) = \frac{Y(z)}{U(z)} = \frac{2.5757\times10^{-6}z^2}{z^2-1.994z+0.9944} \quad m/V$. The linear motor has the following parameters; K_A=0.4021 A/V, K_F=45.6 N/A, M_M=17.7 kg, B_M=18.5 Ns/m and T=0.0005 s.

A PM as described earlier is used to model the hysteresis and inverse compensation is applied to cancel out the dominant nonlinear effect. Thus, after the compensation, the linear part of the actuator plus the additional linear block can be described as follows: $G_{MA}(z) = \frac{Y(z)}{U(z)} = \frac{2.4025z+0.8354}{z^2-0.4158z+0.04326} \quad um/V$.

In the case of designing a controller for a DSA system, it is desirable to control the overall system's position output Y_t by manipulating the input voltage to the linear motor u_c, and the input voltage to the magnetostrictive actuator u_f. Double linear transformation is used to convert the plant models into s domain in the form of a state space representation. The DSA system is described by the following equations:

$$\begin{bmatrix} u_f \\ u_c \end{bmatrix} = \mathbf{B}^{-1}\left[\begin{pmatrix} x_{fd}^{(2)} - \lambda \tilde{x}_{fd}^{(1)} \\ x_{cd}^{(2)} - \lambda \tilde{x}_{cd}^{(1)} \end{pmatrix} - \begin{pmatrix} f_f + \hat{d}_f \\ f_c + \hat{d}_c \end{pmatrix} + \begin{pmatrix} k_f\, sgn\,(s_f) \\ k_c\, sgn\,(s_c) \end{pmatrix} + \begin{pmatrix} k_{sf}s_f \\ k_{sc}s_c \end{pmatrix}\right]$$

where f_i is the system dynamics and $\hat{d}_i$ is the disturbance estimation; s_i is the sliding surface for linear motor and magneto actuator. We treat the model error and parametric uncertainties as disturbance. The interaction between the two actuators is so small that the disturbance acting on the linear motor can be ignored. Therefore, the linear motor is an accurate model and its control law is actually in the form of PD controller. We define the maximum model error after hysteresis cancellation as the boundary of the disturbance acting on the magneto actuator. So, the k_f can be determined experimentally as 0.6.

λ is the desired and achievable bandwidth of the controlled dynamic system. It is a strictly positive constant. For the sliding mode control (SMC), the choice of the value of the parameter is critical. A value of λ_c= 200 and λ_f=600

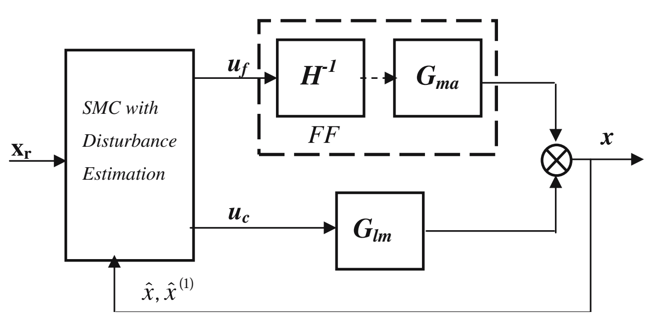
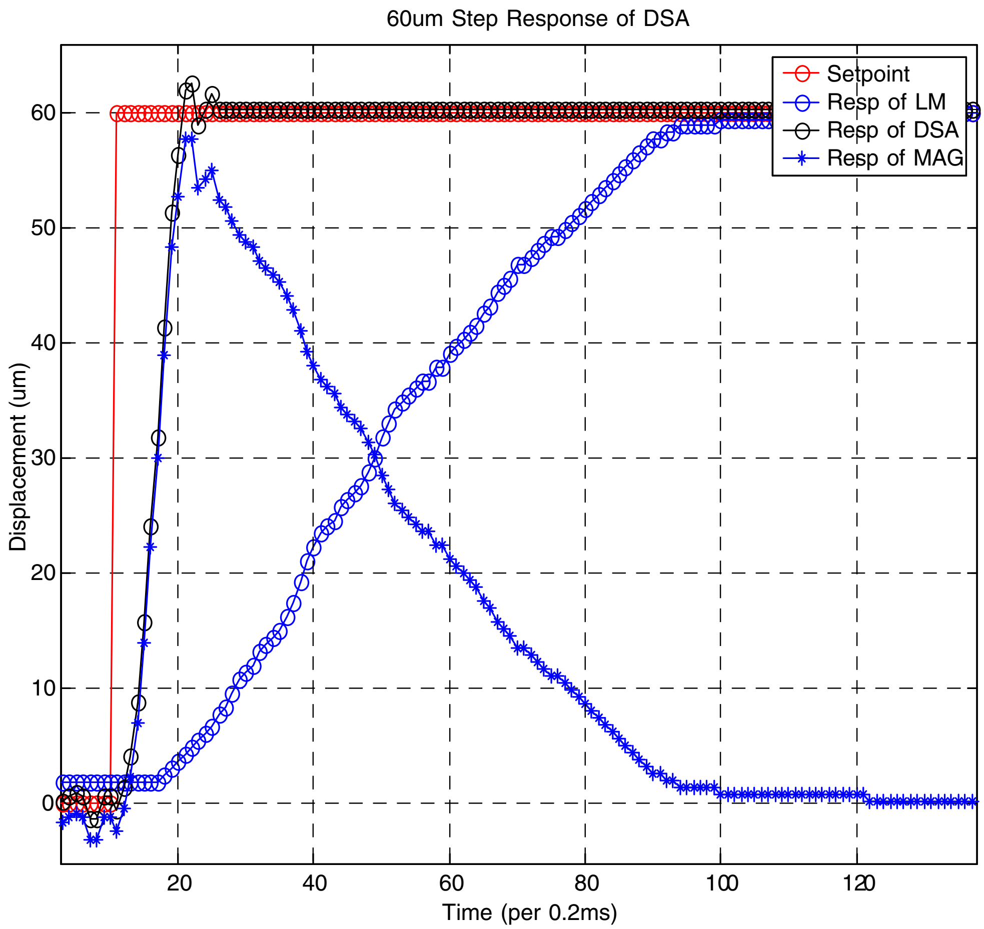
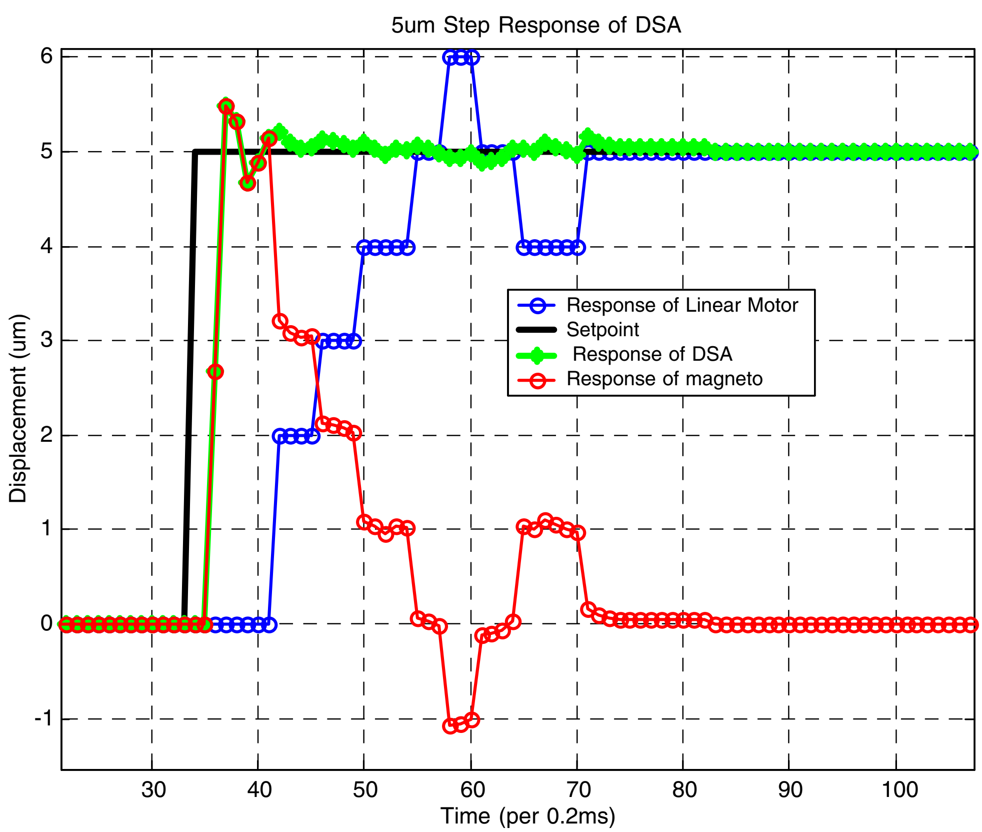


Fig. 7 Block diagram of resultant SMC controller for the DSA system

Fig. 8 Comparison of 60 μm (left_hand) and 5 μm (right_hand) step response between linear motor and the combined system





are determined experimentally for the linear motor and magneto actuator respectively, while taking into consideration three limiting factors [16]:

1. Structural resonant frequencies.
2. Unmodeled time delay: $\lambda \leq 1/3T_d$.
3. Control sampling frequency: $\lambda \leq 0.2f_s$.

It is expected that the fine actuator will respond quickly to shorten the rise time of the whole system or the settling time during the transition period. Then decrease gradually as the contribution of the steady-state output position from the coarse actuator increases. This will provide the fine actuator with the maximum possible room to compensate for errors that the coarse actuator makes due to its slower dynamics and longer dead-time. This is accomplished by setting the individual feedback gain properly to ensure that the two components cooperate as desired. Thus, the controller's block diagram are given below in Fig. 7, in which H^{-1} is the inversed resultant hysteresis.

The performance of the resultant SMC controller was evaluated on the dual-stage actuation system. Two tests have been performed: (1) the case in which the magnitude of the step is 60 and 5 μm. These results are shown in Fig. 8, and; (2) the case where a 90° sharp corner contouring tracking case is evaluated and compared by commanding the DSA to run at 0.1 m/s and then change the velocity direction. This result is shown in Fig. 9. In the case of the step testing, 60 μm step magnitude, which is slightly bigger than the maximum stroke of MA, was chosen to show the cooperation between the two actuators. The MA was activated mainly during the LM's transient period and thus compensated the error caused by the coarse actuator due to its slow dynamics and long dead-time. After reaching steady-state, the LM implements offset-free control and the output of the MA will be backed off to zero or any designated lower value to prepare for the next control move. From the results, it can be seen that the MA has a 0.5 ms dead time and takes 3.2 ms to reach the 60 μm set point at steady-state and LM has 1.6 ms dead time and 15.6 ms rise-time.

From the 5 μm step response result shown at the right hand in Fig. 8, as the linear scale's resolution is 1 micron, the step response of the motor shows staircase-like curve. In this case, a proximity sensor is used to obtain sub-micron level measurement of the overall system's output.

Frequency response and sensitivity function are also measured on the DSA system as shown in Fig. 10. Results show the system has a resonant peak at 220 Hz and 380 Hz bandwidth (−3dB from its static gain).

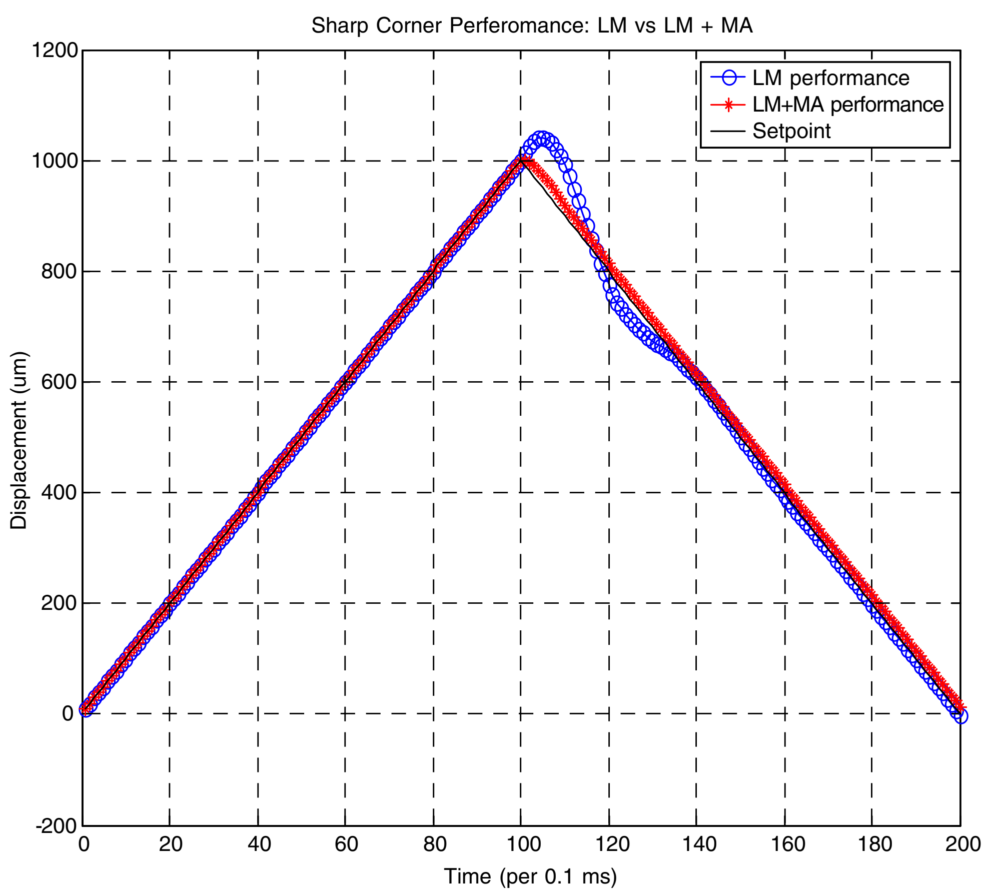
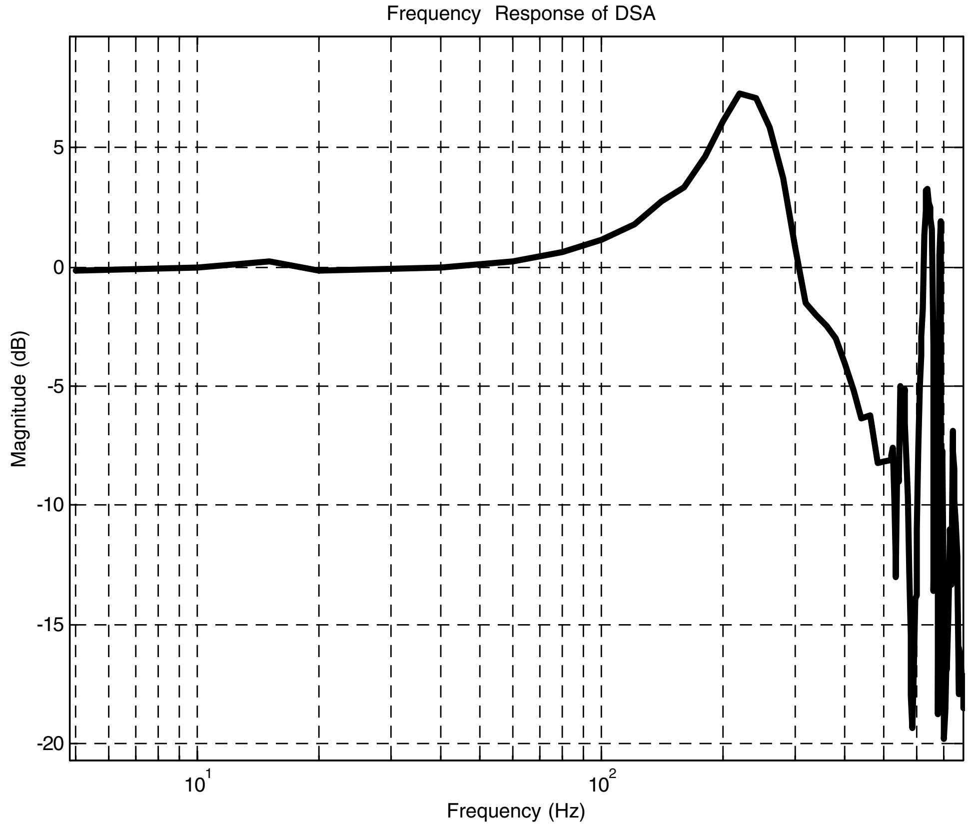
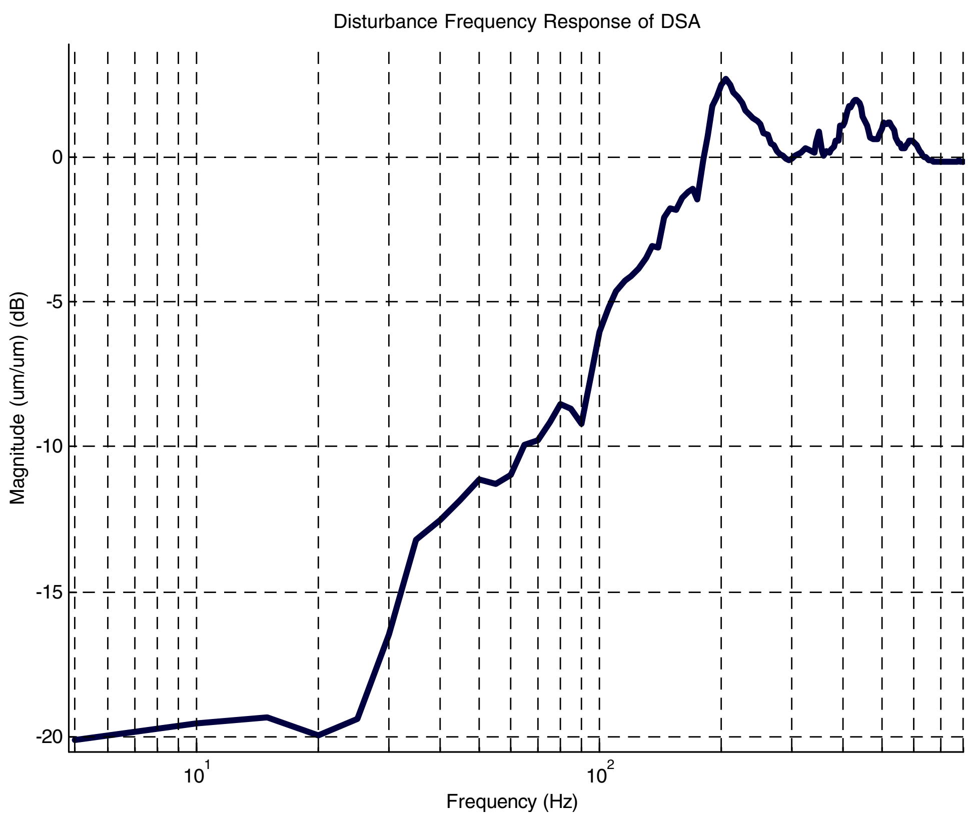


Fig. 9 Comparison of sharp corner contouring between linear motor and the combined system

Fig. 10 Frequency response of the DSA system (left hand) and sensitivity function (right hand) of the DSA system





5 Conclusions

In this paper, a dual stage actuation system was built and tested. A dynamic Preisach model was ultilized to cancel the undesired hysteresis. An SMC controller was designed to deal with uncertainties such as the error in hysteresis cancellation and external disturbance to ensure system stability. The sliding mode controller design resulted in a controller with a simple and closed form structure, most importantly, can ensure the system to be asymptotically stable. Thus this controller structure is suitable for fast servo control applications. Based on feedforward hysteresis cancellation and robust feedback correction, an evaluation on the resultant controller showed that with proper tuning a desirable system behavior to a step response and under sharp corner tracking can be achieved.

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